

LABORATORY RECORD**EXP NO.: 6 IMPLEMENTING FABRIC CHANNELS FOR PRIVACY****AIM**

To create and configure multiple channels in a Hyperledger Fabric network, deploy chaincode on a specific channel, and demonstrate how Fabric channels provide data privacy and isolation between organizations.

PROCEDURE**Step 1: Navigate to the Fabric Test Network**

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Hyperledger Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Verify the Existing Channel

```
peer channel list
```

Step 4: Create a New Private Channel

```
./network.sh createChannel -c privatechannel
```

Step 5: Verify Available Channels

```
peer channel list
```

Step 6: Deploy Chaincode on the Private Channel

```
./network.sh deployCC -c privatechannel -ccn basic \  
-ccl javascript -ccp ../asset-transfer-basic/chaincode-javascript
```

Step 7: Initialize the Ledger

```
peer chaincode invoke -o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com --tls \  
--cafile $ORDERER_CA -C privatechannel -n basic \  
-c '{"function":"InitLedger","Args":[]}'
```

Step 8: Query Assets from the Private Channel

```
peer chaincode query -C privatechannel -n basic \  
-c '{"Args":["GetAllAssets"]}'
```

Step 9: Verify Channel Isolation

```
peer chaincode query -C mychannel -n basic \
  -c '{"Args":["ReadAsset","asset1"]}'
```

Step 10: Stop the Fabric Network

```
./network.sh down
```

PROGRAM

Fabric Channel Creation & Private Chaincode Deployment Shell Script (deploy_private_channel.sh):

```
#!/bin/bash
# Script to create channels and deploy chaincode for data privacy

cd ~/fabric-dev/fabric-samples/test-network

# Start network and create default channel
./network.sh up createChannel -ca

# Create a separate private channel
./network.sh createChannel -c privatechannel

# Verify created channels
peer channel list

# Deploy asset transfer chaincode to privatechannel
./network.sh deployCC -c privatechannel -ccn basic -ccl javascript -ccp ../
asset-transfer-basic/chaincode-javascript

echo "Fabric Channels and Private Chaincode Deployed Successfully."
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -
ca
Creating channel 'mychannel'... done
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Network Started Successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer channel list
Channels peers has joined:
mychannel
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh createChannel -c privatechannel
Channel 'privatechannel' created successfully.

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer channel list
Channels peers has joined:
mychannel
privatechannel
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -c privatechannel -ccn basic -ccl javascript -ccp ../asset-transfer-basic/chaincode-javascript
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode...
Committing Chaincode...
Chaincode committed successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile $ORDERER_CA -C privatechannel -n basic -c '{"function":"InitLedger","Args":[]}'
Ledger Initialized Successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C privatechannel -n basic -c '{"Args":["GetAllAssets"]}'
[
  { "ID": "asset1", "Owner": "Tomoko", "Color": "Blue" },
  { "ID": "asset2", "Owner": "Brad", "Color": "Red" }
]

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset1"]}'
Error: endorsement failure during query. response status 500: chaincode 'basic' not found on channel 'mychannel'
Data stored in privatechannel is not accessible through mychannel. Channel privacy verified successfully.

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Network cleaned up successfully
```

RESULT

The Hyperledger Fabric network was successfully configured with multiple channels. A private channel was created, chaincode was deployed, and ledger transactions were executed successfully. The experiment demonstrated that channel members can access only the data belonging to their respective channels, ensuring privacy and confidentiality in a permissioned blockchain network.